

What is a gem?

A gem is generally defined as any mineral that is highly prized for its beauty, durability, and rarity, is used for personal adornment, and has been enhanced in some manner by altering its shape – usually by cutting and polishing. A wider definition includes a few rocks, such as obsidian, and a few organic substances, such as

amber (a fossilized resin). By far the majority of gems, however, are cut from the crystals of minerals. Precious metals are not considered to be gems, nor are items carved from minerals but not used for personal adornment, such as figurines, bowls, or vases.

Beauty

The first quality a gem must possess is that of beauty. This is subjective: some may prize a gem's interplay of light and colour, while others may first be drawn to a gem's intricate cut. With an almost endless combination of colour, shape, and fire (play of light), gemstones are capable of a range of aesthetic styles.



Blue sapphire, prized for its beauty

Durability

Hardness or toughness in a gem is a desirable quality, suggesting enduring value. Some gems require care to prolong their longevity. Certain gems resist chipping or scratching, but fade after long exposure to direct light; dry environments may cause some to crack, while others are susceptible to damage from acids.



Diamond, the hardest gemstone

Synthetic gems

Synthetic gems are identical to natural minerals physically, chemically, and optically, but are made in a laboratory. The two main ways to create them are from melt or solution. In production from melt, a powdered material, chemically equivalent to a natural mineral, is melted at high heat, then manipulated to solidify in a crystalline form. Production from solution involves dissolving one set of materials in a solution of different materials, again using high heat, then manipulating the solution so it precipitates into crystalline form. In both methods, crystals form on a seed crystal as the temperature is lowered.

Natural gemstones for comparison



Opal

Synthetic examples of gemstones



Synthetic opal

Large patches of colour with snakeskin effect seen up close



Diamond



Synthetic diamond

Unusually flawless interior rarely found in nature

Rarity

A gem may be rare for a number of reasons. The gem material itself may be rare, such as emerald, or a more common material may exhibit an unusual colour or clarity. Some particularly soft or fragile stones are rare in cut form, as they require the work of highly skilled lapidaries.



Taaffeite, noted for its rarity

Other considerations

The desirability of a gem can depend on factors besides beauty, rarity, and durability. Gems may be symbolic of power, such as those mounted in crowns, or valued for their history or circumstances of origin. They may also be prized for their connection to astrology or mysticism, for their geological associations, or as fashion items.



Tanzanite, both rare and difficult to cut



Emerald



Synthetic emerald

No visible internal flaws, unlike real emeralds